V' of V_1' in V which contains $V_2 \oplus V_3$. This gives us the decomposition

$$\begin{tikzcd}
 & V & \\
V_1 \arrow[r, hook] & \bullet & V_3 \arrow[l, hook'] \\
 & V_2 \arrow[u, hook] &
\end{tikzcd}$$

$$= \quad \begin{tikzcd}
V_1' \arrow[r, "\sim"] & V_1' & 0 \arrow[l] \\
 & 0 \arrow[u] &
\end{tikzcd} \quad \oplus \quad \begin{tikzcd}
Y \arrow[r, hook] & V' & V_3 \arrow[l, hook'] \\
 & V_2 \arrow[u, hook] &
\end{tikzcd}.$$

The first of these summands is a multiple of

$$\begin{tikzcd}
1 \arrow[r, "\sim"] & 1 & 0 \arrow[l] \\
 & 0 \arrow[u] &
\end{tikzcd}.$$

By splitting these away, we get to a situation where $V_1 \subseteq V_2 \oplus V_3$. Similarly, we can split away objects of the type

$$\begin{tikzcd}
0 \arrow[r] & 1 & 0 \arrow[l] \\
 & 1 \arrow[u, squiggly] &
\end{tikzcd} \quad \text{and} \quad \begin{tikzcd}
0 \arrow[r] & 1 & 1 \arrow[l, "\sim"'] \\
 & 0 \arrow[u] &
\end{tikzcd}$$

to reach a situation in which the following conditions hold:

(1) $V_1 + V_2 + V_3 = V$.

(2) $V_1 \cap V_2 = 0, \quad V_1 \cap V_3 = 0, \quad V_2 \cap V_3 = 0$.

(3) $V_1 \subseteq V_2 \oplus V_3, \quad V_2 \subseteq V_1 \oplus V_3, \quad V_3 \subseteq V_1 \oplus V_2$.

But this implies that

$$V_1 \oplus V_2 = V_1 \oplus V_3 = V_2 \oplus V_3 = V.$$

So we get

$$\dim V_1 = \dim V_2 = \dim V_3 = n$$